This stems from the fact that for a given cooling arrangement, the electric and magnetic loadings of machines of different types will be more or less the same. The torque per unit length therefore depends first and foremost on the square of the diameter, so motors of roughly the same diameter and length can be expected to produce roughly the same torque.

1.9.3 Power per unit volume and efficiency—Importance of speed

Output power per unit volume is directly proportional to speed.

Low-speed motors are unattractive for most applications because they are large, and therefore expensive. It is usually better to use a high-speed motor with mechanical speed reduction. For example, a direct drive motor for a portable electric screwdriver would be an absurd proposition. On the other hand the reliability and inefficiency of gearboxes may sometimes outweigh the size argument, especially in high-power applications.

The efficiency of a motor improves with speed.

For a given torque, power output usually rises in direct proportion to speed, while electrical losses tend to rise less rapidly, so that efficiency rises with speed.

1.9.4 Size effects—Specific torque and efficiency

Large motors have a higher specific torque (torque per unit volume) and are more efficient than small ones.

In large motors the specific electric loading is normally much higher than in small ones, and the specific magnetic loading is somewhat higher. These two factors combine to give the higher specific torque.

Very small motors are inherently very inefficient (e.g. 1% in a wrist-watch), whereas motors of over say 100 kW have efficiencies above 96%. The reasons for this scale effect are complex, but stem from the fact that the resistance volt-drop term can be made relatively small in large electromagnetic devices, whereas in small ones the resistance becomes the dominant term.

1.9.5 Rated voltage

A motor can be provided to suit any voltage.

Within limits it is possible to rewind a motor for a different voltage without affecting its performance. A 200 V, 10 A motor could be rewound for 100 V, 20 A simply by using half as many turns per coil of wire having twice the cross-sectional area, the m.m.f. remaining the same. The total amounts of active material, and hence the performance, would be the same. This argument breaks down if pushed too far of course: a very small motor originally wound for 100 V would almost certainly require a larger frame if required to operate at 690 V, because of the additional space required for the higher-grade insulation.

1.9.6 Short-term overload

Most motors can be overloaded for short periods without damage.

The continuous electric loading (i.e. the current) cannot be exceeded without overheating and damaging the insulation, but if the motor has been running with reduced current for some time, it is permissible for the current (and hence the torque) to be much greater than normal for a short period of time. The principal factors which influence the magnitude and duration of the permissible overload are the thermal time-constant (which governs the rate of rise of temperature) and the previous pattern of operation. Thermal time constants range from a few seconds for small motors to many minutes or even hours for large ones. Operating patterns are obviously very variable, so rather than rely on a particular pattern being followed, it is usual for motors to be provided with over-temperature protective devices (e.g. thermistors) which trigger an alarm and/or trip the supply if the safe temperature is exceeded.

Motors can of course be designed for specific applications with known duty cycles, where the thermal and electric loading can be accommodated. This is discussed further in Chapter 11.

1.10 Review questions

(1) The current in a coil with 250 turns is 8 A. Calculate the m.m.f.

(2) The coil in Q1 is used in a magnetic circuit with a uniform cross-section made of good-quality magnetic steel and with a 2 mm air-gap. Estimate the flux density in the air-gap, and in the iron. ($\mu_0 = 4 \times 10^{-7}\ \mathrm{Hm^{-1}}$)

How would the answers change if the cross-sectional area of the magnetic circuit was doubled, with all other parameters remaining the same?

(3) A magnetic circuit of uniform cross-sectional area has two air-gaps of 0.5 and 1 mm respectively in series. The exciting winding provides an m.m.f. of 1200 Amp-turns. Estimate the m.m.f. across each of the air-gaps, and the flux density.

(4) The rotor of a d.c. motor had an original diameter of 30 cm and an air-gap under the poles of 2 mm. During refurbishment the rotor diameter was accidentally reground and was then undersize by 0.5 mm. Estimate by how much the field m.m.f. would have to be increased to restore normal performance. How might the extra m.m.f. be provided?

(5) Calculate the electromagnetic force on:

- **(a)** a single conductor of length 25 cm, carrying a current of 4A, exposed to a magnetic flux density of 0.8 T perpendicular to its length.
- **(b)** a coil-side consisting of 20 wires of length 25 cm, each carrying a current of 2 A, exposed to a magnetic flux density of 0.8 T perpendicular to its length.

(6) Estimate the torque produced by one of the early machines illustrated in Fig. 1.12 given the following:- Mean air-gap flux density under pole-face = 0.4 T; pole-arc as a percentage of total circumference = 75%; active length of rotor = 50 cm; rotor diameter = 30 cm; inner diameter of stator pole = 32 cm; total number of rotor conductors = 120; current in each rotor conductor = 50 A.

(7) If the field coils of a motor are rewound to operate from 220 V instead of 110 V, how will the new winding compare with the old in terms of number of turns, wire diameter, power consumption and physical size?

(8) A catalogue of DIY power tools indicates that most of them are available in 240 V or 110 V versions. What differences would you expect in terms of appearance, size, weight and performance?

(9) Given that the field windings of a motor do not contribute to the mechanical output power, why do they consume power continuously?

(10) Explain briefly why low-speed electrical drives often employ a high-speed motor and some form of mechanical speed reduction, rather than a direct-drive motor.

Answers to the review questions are given in the Appendix.

only half what it was when the motor was at full speed. This is because at high slips, the leakage fluxes assume a much greater importance than under normal low-slip conditions. The simple arguments we have advanced to predict torque would therefore need to be modified to take account of the reduction of main flux if we wanted to use them quantitatively at high slips. There is no need for us to do this explicitly, but it will be reflected in any subsequent curves portraying typical torque-speed curves for real motors. Such curves are of course used when selecting a motor to run directly from the utility supply, since they provide the easiest means of checking whether the starting and run-up torque is adequate for the job in hand. Fortunately, we will see in Chapter 7 that when the motor is fed from an inverter, we can avoid the undesirable effects of high-slip operation, and guarantee that the flux is at its optimum value at all times.

5.5 Stator current-speed characteristics

To conclude this chapter we will look at how the stator current behaves, remembering that we are assuming that the machine is directly connected to a utility supply of fixed voltage and frequency. Under these conditions the maximum current likely to be demanded and the power factor at various loads are important matters that influence the running cost.

In the previous section, we argued that as the slip increased, and the rotor did more mechanical work, the stator current increased. Since the extra current is associated with the supply of real (i.e. mechanical output) power (as distinct from the original magnetising current which was seen to be reactive), this additional 'work' component of current is more or less in phase with the supply voltage, as shown in the phasor diagrams, Fig. 5.23.

The resultant stator current is the sum of the magnetising current, which is present all the time, and the load component, which increases with the slip.

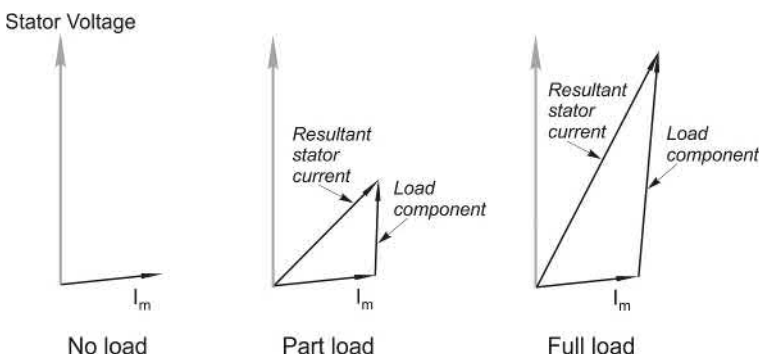

FIG. 5.23 Phasor diagrams showing stator current at no-load, part-load and full-load. The resultant current in each case is the sum of the no-load (magnetising) current and the load component.

We can see that as the load increases, the resultant stator current also increases, and moves more nearly into phase with the voltage. But because the magnetising current is appreciable, the difference in magnitude between no-load and full-load currents may not be all that great. (This is in sharp contrast to the d.c. motor, where the no-load current in the armature is very small in comparison with the full-load current. Note however that in the d.c. motor, the excitation (flux) is provided by a separate field circuit, whereas in the induction motor the stator winding furnishes both the excitation and the work currents. If we consider the behaviour of the work components of current only, both types of machine look very similar.)

The simple ideas behind Fig. 5.23 are based on an approximation, so we cannot push them too far: they are fairly close to the truth for the normal operating region, but break down at higher slips, where the rotor and stator leakage reactances become significant. A typical current locus over the whole range of slips for a cage motor is shown in Fig. 5.24. We note that the power factor is poor when the motor is lightly loaded, and becomes worse again at high slips, and also that the current at standstill (i.e. the 'starting' current) is perhaps five times the full-load value.

Very high currents when started direct-on-line are one of the worst features of the cage induction motor. They not only cause unwelcome volt-drops in the supply system, but also call for heavier switchgear than would be needed to cope with full-load conditions. Unfortunately, for reasons discussed earlier, the high starting currents are not accompanied by high starting torques, as we can see from Fig. 5.25, which shows current and torque as functions of slip for a general-purpose cage motor.

We note that the torque per ampere of current drawn from the utility supply is typically very low at start up, and only reaches a respectable value in the normal operating region, i.e. when the slip is small. This matter is explored further in Chapter 6.

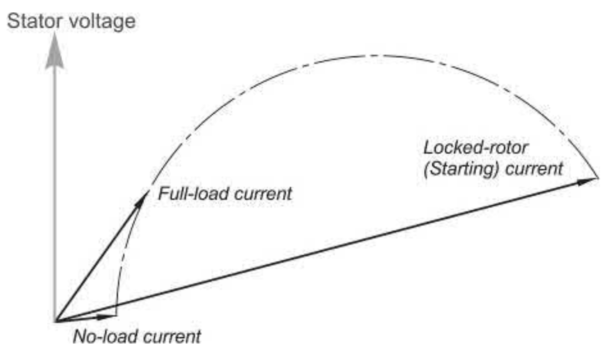

FIG. 5.24 Phasor diagram showing the locus of stator current over the full range of speeds from no-load (full speed) down to the locked-rotor (starting) condition.

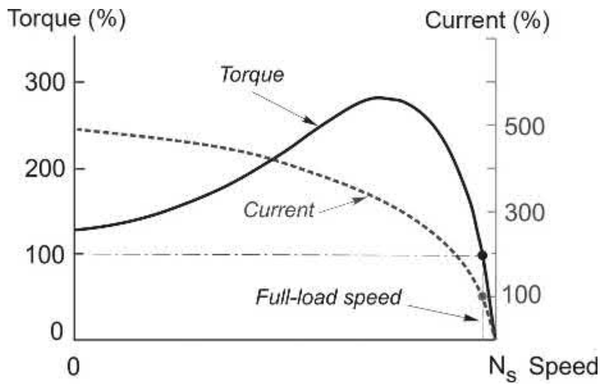

FIG. 5.25 Typical torque-speed and current-speed curves for a cage induction motor. The torque and current axes are scaled so that 100% represents the continuously-rated (full-load) value.

5.6 Review questions

(1) The nameplate of a standard 50 Hz induction motor quotes full-load speed as 2950 rev/min. Find the pole-number and the rated slip.

(2) A four-pole, 60 Hz induction motor runs with a slip of 4%. Find:
- **(a)** the speed;
- **(b)** the rotor frequency;
- **(c)** the speed of rotation of the rotor current wave relative to the rotor surface;
- **(d)** the speed of rotation of the rotor current wave relative to the stator surface.

(3) An induction motor designed for operation from 440 V is supplied at 380 V instead.

What effect will the reduced voltage have on the following:
- **(a)** the synchronous speed;
- **(b)** the magnitude of the air-gap flux;
- **(c)** the induced current in the rotor when running at rated speed;
- **(d)** the torque produced at rated speed.

(4) The rotor from a six-pole induction motor is to be used with a four-pole stator having the same bore and length as the six-pole stator. What modifications would be required to the rotor if it was
- **(a)** a squirrel-cage type, and
- **(b)** a wound-rotor type?

(5) A 440 V, 60 Hz induction motor is to be used on a 50 Hz supply. What voltage should be used?

(6) The stator of a 220 V induction motor is to be rewound for operation from a 440 V supply. The original coils each had 15 turns of 1 mm diameter wire. Estimate the number of turns and diameter of wire for the new stator coils.

second shock—albeit of lesser severity—to the supply system and to the load. For star/delta switching to be possible both ends of each phase of the motor windings must be brought out to the terminal box. This requirement is met in the majority of motors, except small ones, which are usually permanently connected in delta.

With a star/delta starter the current drawn from the supply is approximately one third of that drawn in a direct-on-line start, which is very welcome, but at the same time the starting torque is also reduced to one third of its direct-on-line value. Naturally we need to ensure that the reduced torque will be sufficient to accelerate the load, and bring it up to a speed at which it can be switched to delta without an excessive jump in the current.

Various methods are used to detect when to switch from star to delta. Historically, in manual starters, the changeover is determined by the operator watching the ammeter until the current has dropped to a low level, or listening to the sound of the motor until the speed becomes steady. Automatic versions are similar in that they detect either falling current, or speed rising to a threshold level, or, where the load is always the same, they operate after a pre-set time.

6.2.3 Autotransformer starter

A three-phase autotransformer is usually used where star-delta starting provides insufficient starting torque. Each phase of an autotransformer consists of a single winding on a laminated core. The incoming supply is connected across the ends of the coils, and one or more tapping points (or a sliding contact) provide a reduced voltage output, as shown in Fig. 6.3.

The motor is first connected to the reduced voltage output, and when the current has fallen to the running value, the motor leads are switched over to the full voltage.

If the reduced voltage is chosen so that a fraction α of the line voltage is used to start the motor, the starting torque is reduced to approximately α^2 times its direct-on-line value, and the current drawn from the supply is also reduced to α^2

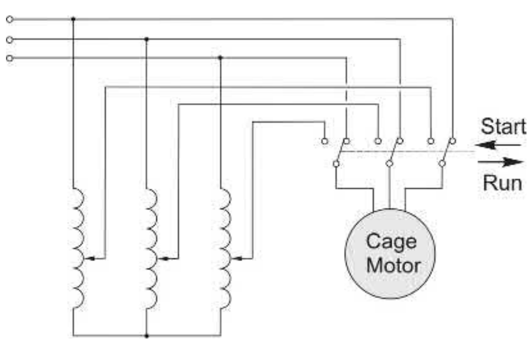

FIG. 6.3 Autotransformer starter for cage induction motor.

times its direct value. As with the star/delta starter, the torque per ampere of supply current is the same as for a direct start.

6.2.4 Resistance or reactance starter

By inserting three resistors or inductors of appropriate value in series with the motor, the starting current can be reduced by any desired extent, but only at the expense of a disproportionate reduction in starting torque.

For example if the current is reduced to half its direct-on-line value, the motor voltage will be halved, so the torque (which is proportional to the square of the voltage—see later) will be reduced to only 25% of its direct-on-line value. This approach is thus less attractive in terms of torque per ampere of supply current than the star/delta method. One attractive feature, however, is that as the motor speed increases and its effective impedance rises, the volt-drop across the extra impedance reduces, so the motor voltage rises progressively with the speed, thereby giving more torque. When the motor is up to speed, the added impedance is shorted-out by means of a contactor.

6.2.5 Solid-state soft starting

This method is now widely used. It provides a smooth build-up of current and torque, the maximum current and acceleration time are easily adjusted, and it is particularly valuable where the load must not be subjected to sudden jerks. The only real drawback over conventional starters is that the utility supply currents during run-up are not sinusoidal, which can lead to interference with other equipment on the same supply.

The most widely-used arrangement comprises three pairs of back-to-back thyristors (or triacs) connected in series with the three supply lines, as shown in Fig. 6.4A.

Each thyristor is fired once per half-cycle, the firing being synchronised with the utility supply and the firing angle being variable so that each pair conducts for a controllable proportion of a cycle. Typical current waveforms are shown in Fig. 6.4B: they are clearly not sinusoidal but the motor will tolerate them quite happily.

A wide variety of control philosophies can be found, with the degree of complexity and sophistication being reflected in the price. The cheapest open-loop systems simply alter the firing angle linearly with time, so that the voltage applied to the motor increases as it accelerates. The ‘ramp-time’ can be set by trial and error to give an acceptable start, i.e. one in which the maximum allowable current from the supply is not exceeded at any stage. This approach is reasonably satisfactory when the load remains the same, but requires resetting each time the load changes. Loads with high static friction are a problem because nothing happens for the first part of the ramp, during which time the motor torque is insufficient to move the load. When the load finally moves,

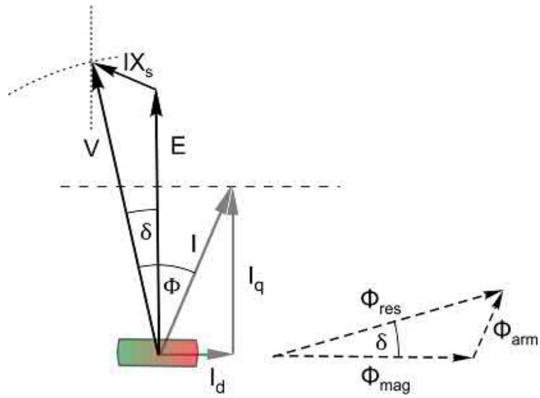

FIG. 9.19 Phasor diagram for the permanent magnet synchronous motor.

green regions within the stylised rotor outline indicate the poles of the magnet, and thus define the direct axis (horizontal) in the diagram. The magnet flux (Φ_{mag}) remains constant, so we choose it as our horizontal reference throughout this and the following sections.

Fig. 9.19 is similar to Fig. 9.16, but we have chosen to add the flux phasors in order to lay the groundwork for the later discussion of field-oriented control. As explained below, the flux and voltage triangles are similar, and each flux is proportional to the corresponding m.m.f. (see Fig. 9.6): for example, the rotor flux (φ_{mag}) corresponds with the rotor m.m.f. (F_R in Fig. 9.6).

E is the open circuit e.m.f. produced by the magnet flux (φ_{mag}): it is proportional to the magnet flux and the speed, which is proportional to the stator frequency, ω. Once we specify the frequency the magnitude of E is known, so we can start the left hand side of the diagram with E drawn vertically because, as previously discussed, the induced e.m.f. leads the magnet flux by 90°.

φ_{arm} is the flux that would be set up if the armature (stator) current existed alone. We are assuming no saliency, so the magnitude and direction of this flux depend only on the magnitude and phase of the stator current (I).

φ_{res} is the resultant ('the') flux. With resistance ignored, this flux is effectively determined by the applied voltage and frequency, as we have seen several times earlier in this book, so in the phasor diagram, the magnitude of this flux is proportional to, and in quadrature with, V.

As usual, the self-induced e.m.f. from the stator flux is modelled by means of the voltage across the effective stator inductive reactance, X_s, so in the phasor diagram the voltage IX_s leads the current by 90°.

V is the applied stator voltage, here regarded as an independent variable, and I is the resulting stator current. The stator phasor equation is $V = E + IX_s$ (the terms being vectors, of course), or $V = E + jIX_s$ for those who prefer the complex notation.

The other independent variable is the load torque, which is our next consideration in developing the diagram. We dealt with the same matter in relation to Fig. 9.16, where the utility supply voltage was constant. But now the voltage is not fixed, so we will take a different approach that focuses instead on how we need to control the current in order to achieve the required torque in the most efficient way. This will assist us in understanding the strategy employed in the torque control loop of a PM motor drive, which we look at later in this chapter.

In the steady state, the motor torque must equal the load torque. The motor torque is proportional to the product of the current (I), the magnet flux (φ_{mag}), and the sine of the angle between them (δ), or in other words to the product of the magnet strength and the component of current at right angles to the magnet flux. Hence the load torque determines the torque component of the stator current (I_q), as shown in Fig. 9.19. The locus of the current I is then the horizontal dashed line, and the locus of V is the vertical dotted line as shown in Fig. 9.19. So when we finally specify the magnitude of V (shown by the dotted arc), the intersection of the arc and vertical line fixes the phase of the stator current, thereby specifying the direct axis component I_d and finalising the diagram.

The current component in phase with E is the useful or torque component I_q, while the component in phase with the magnet flux is the flux component I_d. The area of the flux triangle is proportional to the product of I_q and φ_{mag}, and thus the area provides an immediate visual indication of the torque.

If we adjust V so that the stator current is in phase with E, (i.e. the flux component I_d is zero), we get the maximum torque per ampere of stator current, and thus the minimum stator copper loss for that particular torque: if iron loss is ignored, this would be the most efficient condition.

It is important to stress that the conclusion that we have reached—that the torque is proportional to the quadrature component of current - is not restricted to the steady-state condition that we have assumed so far, and is in fact the basis of the field-oriented control system that we will discuss in Section 9.6.

Alternative expressions for the power per phase can be derived from the geometry of the phasor diagram with the aid of a few construction lines. The first is identical to that obtained for the excited rotor motor, i.e. $P = \frac{EV}{Xs}\sin\delta$, while the second, which is more appropriate when the current is being controlled directly, is $P = EI_q$. The second equation is of exactly the same form as we found for the d.c. machine in Chapter 3.

Before we move on, we should note that because we looked at an example where the applied voltage is just a bit larger than E, the current turned out to be of modest amplitude and at a reasonable power-factor angle. However, if we had specified a much higher V, we would find that the current would have had a much larger lagging flux component (with the same torque component, determined by the load), and a much worse power factor. And conversely, a much smaller applied voltage would result in a large leading power-factor current. This behaviour is in line with what we have already seen, and neither condition is desirable because of the increased copper loss.